郭麗[illegible] 11370044

14.4

10. Marriage Rates Half of adults today are married compared with the all-time high of 72% in 1960.[9] Do marriage rates vary by educational level? A sample of 300 adults in each of three educational categories provided the following information.

	Married		
Educational Level	Yes	No	total
Bachelors or higher	187	113	300
Some college	162	138	300
High school or less	149	151	300
total	498	402	900

Test to determine whether marriage rates differ significantly among adults in these educational levels using $\alpha = .01$. How would you summarize your results?

Ho: Marriage rates are the same for all educational levels
Ha: Marriage rates differ among educational levels

$E_{yes} = \frac{300 \times 498}{900} = 166$

$E_{no} = \frac{300 \times 402}{900} = 134$

tstat: $X^2 = \Sigma \frac{(O-E)^2}{E}$

$X^2 = \frac{(187-166)^2}{166} + \frac{(162-166)^2}{166} + \frac{(149-166)^2}{166} + \frac{(113-134)^2}{134} + \frac{(138-134)^2}{134} + \frac{(151-134)^2}{134}$

$= 2.657 + 0.096 + 1.741 + 3.291 + 0.119 + 2.157$

$= 10.06$

$df = (r-1)(c-1) = (3-1)(2-1) = 2$

$\alpha = 0.01,\ df = 2 \rightarrow X^2_{0.01,2} = 9.210$

RR: $X^2 > 9.210$

$10.06 > 9.210$ (Reject Ho)

⟶ There is sufficient evidence at 0.1 significance level to conclude that marriage rates differs significantly among educational levels.

⟶ People with higher education levels appear more likely to be married

12. How Big Is the Household? A local chamber of commerce surveyed 120 households in their city—40 in each of three types of residence (apartment, duplex, or single residence)—and recorded the number of family members in each of the households. The data are shown in the table.

	Type of Residence			
Family Members	Apartment	Duplex	Single Residence	total
1	8	20	1	29
2	16	8	9	33
3	10	10	14	34
4 or More	6	2	16	24
total	40	40	40	120

Is there a significant difference in the family size distributions for the three types of residence? Test using $\alpha = .01$. If there are significant differences, describe the nature of these differences.

Ho: The family size distributions are the same for all residence types
Ha: At least one residence type has a different family size distribution

Expected Table

Members	Apart	Duplex	Single Residence	
1	9.667	9.667	9.667	$\frac{29 \times 40}{120} = 9.667$
2	11	11	11	$\frac{33 \times 40}{120} = 11$
3	11.333	11.333	11.333	$\frac{34 \times 40}{120} = 11.333$
4+	8	8	8	$\frac{24 \times 40}{120} = 8$

tstat: $X^2 = \Sigma \frac{(O-E)^2}{E} \approx 36.49$

$df = (r-1)(c-1) = (4-1)(3-1) = 6$

$\alpha = 0.01,\ df = 6 \rightarrow X^2_{0.01,6} = 16.812$

RR: $X^2 > 16.812$

$36.49 > 16.812$ (Reject Ho)

⟶ There is sufficient evidence at 0.01 significance level to conclude that family size distributions differ among the three residence types.

Interpretation

- Duplexes have many more 1-person household than expected
- Single residences have many more large families (4/more) than expected
- Apartments are more concentrated around 2-3 family members

So, residence types appears related to size of household.

15.1

12. Dissolved Oxygen Content Dissolved oxygen content is a measure of the ability of a river, lake, or stream to support aquatic life, with high levels being better. A pollution-control inspector who suspected that a river community was releasing semitreated sewage into a river, randomly selected five specimens of river water at a location above the town and another five below. These are the dissolved oxygen readings (in parts per million):

Above Town	4.8	5.2	5.0	4.9	5.1
Below Town	5.0	4.7	4.9	4.8	4.9

a. Use a one-tailed Wilcoxon rank sum test with $\alpha = .05$ to confirm or refute the theory.

b. Use a Student's t-test (with $\alpha = .05$) to analyze the data. Compare the conclusion reached in part a.

a) Ho: The two populations are the same
Ha: $M_A > M_B$

Value	Group	Rank
4.7	B	1
4.8	A	2.5
4.8	B	2.5
4.9	A	5
4.9	B	5
4.9	B	5
5.0	A	7.5
5.0	B	7.5
5.1	A	9
5.2	A	10

$T_A = 2.5 + 5 + 7.5 + 9 + 10 = 34$

$T_A^* = n_1(n_1 + n_2 + 1) - T_A$

$= 5(5+5+1) - 34 = 55 - 34 = 21$

$T = \min(T_A, T_A^*) = \min(34, 21) = 21$

$T_c = 19 \rightarrow T = 21 > 19$ (Fail to reject Ho)

⟶ There is no sufficient evidence at $\alpha = 0.05$ to conclude that dissolved oxygen above town is higher.

b) $\bar{x}_A = \frac{4.8 + 5.2 + 5.0 + 4.9 + 5.1}{5} = 5$

$\bar{x}_B = \frac{5.0 + 4.7 + 4.9 + 4.8 + 4.9}{5} = 4.86$

$\bar{x}_A - \bar{x}_B = 0.14$

$S_1^2 = \frac{0.10}{5-1} = 0.025 \qquad S_2^2 = \frac{0.052}{5-1} = 0.013$

$Sp^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1 + n_2 - 2}$

$= \frac{4(0.025) + 4(0.013)}{8} = 0.019$

$t = \frac{\bar{x}_A - \bar{x}_B}{\sqrt{Sp^2\left(\frac{1}{n_A} + \frac{1}{n_B}\right)}} = \frac{0.14}{\sqrt{0.0076}} = \frac{0.14}{0.08718} = 1.61$

$t_{8,0.05} = 1.86$

$t = 1.61 < 1.86$ (Failed to reject Ho)

⟶ Both tests give the same conclusion

13. Eye Movement In an investigation of the visual scanning behavior of deaf children, measurements of eye movement were taken on nine deaf and nine hearing children. The table gives the eye-movement rates and their ranks (in parentheses). Does it appear that the distributions of eye-movement rates for deaf children and hearing children differ?

	Deaf Children	Hearing Children
	2.75 (15)	.89 (1)
	2.14 (11)	1.43 (7)
	3.23 (18)	1.06 (4)
	2.07 (10)	1.01 (3)
	2.49 (14)	.94 (2)
	2.18 (12)	1.79 (8)
	3.16 (17)	1.12 (5.5)
	2.93 (16)	2.01 (9)
	2.20 (13)	1.12 (5.5)
Rank Sum	126	45

Ho: The two distributions are the same
Ha: The two distributions differ

$n_1 = n_2 = 9$

$T = \min(T, T^*) = (126, 45)$
$= 45$

$T_c = 63$

$T = 45 < 63$ (Reject Ho)

⟶ There is sufficient evidence to conclude that the distributions of eye movement rates for deaf & hearing children are different.

15.2

Two Simple Examples *Use the sign test to compare two populations for significant differences for the paired data in Exercises 4–5. State the null and alternative hypotheses to be tested. Determine an appropriate rejection region with $\alpha \le .10$. Calculate the observed value of the test statistic and present your conclusion.*

4.

	Pair						
Population	1	2	3	4	5	6	7
1	8.9	8.1	9.3	7.7	10.4	8.3	7.4
2	8.8	7.4	9.0	7.8	9.9	8.1	6.9
	+	+	+	−	+	+	+

$p = P(\text{Population 1} > \text{Population 2})$

$H_0: p = 0.5$

$H_a: p \neq 0.5$

$n = 7, \; x^* = 6$

$X \sim Bin(7, 0.5) \quad \alpha \le 0.10$

$P(X \le 1) = 0.062$ $\quad$ $P(X \le 0) = 0.008$

p-val $= 2(0.062) = 0.124$ $\quad$ p-val $= 2(0.008) = 0.016$

RR : $x = 0$ or $x = 7$

$x = 6 \neq 0$, $6 \neq 7$ (Failed to reject H_0)

→ There is no sufficient evidence to conclude that the two populations are significantly different

DATA SET DS1507

11. Gourmet Cooking Two chefs, A and B, rated 22 meals on a scale of 1 – 10. The data are shown in the table. Do the data provide sufficient evidence to indicate that one of the chefs tends to give higher ratings than the other? Test by using the sign test with a value of α near .05.

Meal	A	B	Meal	A	B
1	6	8 −	12	8	5 +
2	4	5 −	13	4	2 +
3	7	4 +	14	3	3 0
4	8	7 +	15	6	8 −
5	2	3 −	16	9	10 −
6	7	4 +	17	9	8 +
7	9	9 0	18	4	6 −
8	7	8 −	19	4	3 +
9	2	5 −	20	5	4 +
10	4	3 +	21	3	2 +
11	6	9 −	22	5	3 +

$n = 20$

$(-) = 9$

$(+) = 11$

a. Use the binomial tables in Appendix I to find the exact rejection region for the test.

b. Use the large-sample z statistic. (NOTE: Although the large-sample approximation is suggested for $n \ge 25$, it works fairly well for values of n as small as 15.)

c. Compare the results of parts a and b.

$H_0: p = 0.5$

$H_a: p \neq 0.5$

a) $X \sim Bin(20, 0.5)$ $\quad$ $20 - 5 = 15$

$P(X \le 5) = 0.0207$

p-val $= 2(0.0207) = 0.0414$

RR : $x \le 5$ or $x \ge 15$

$x^* = 11 \not\le 5$, $11 \not\ge 15$ (Failed to reject H_0)

b) $z = \dfrac{x - 0.5n}{0.5\sqrt{n}} = \dfrac{11 - 0.5(20)}{0.5\sqrt{20}} = \dfrac{11 - 10}{2.236} = 0.447$

$z_{0.025} = 1.96$

RR : $|z| > 1.96$ → $0.447 < 1.96$ (Fail to reject H_0)

c) Both methods give same conclusion

There is no sufficient evidence to conclude that one chef tends to give higher rating than others

DATA SET DS1508

13. Recovery Rates Clinical data concerning the effectiveness of two drugs in treating a particular condition (as measured by recovery in 7 days or less) were collected from 10 hospitals. You want to know whether the data present sufficient evidence to indicate a higher recovery rate for one of the two drugs.

a. Test using the sign test. Choose your rejection region so that α is near .05.

b. Why might it be inappropriate to use the Student's t-test in analyzing the data?

	Drug A		
Hospital	Number in Group	Number Recovered (7 days or less)	Percentage Recovered
1	84	63	75.0
2	63	44	69.8
3	56	48	85.7
4	77	57	74.0
5	29	20	69.0
6	48	40	83.3
7	61	42	68.9
8	45	35	77.8
9	79	57	72.2
10	62	48	77.4

	Drug B		
Hospital	Number in Group	Number Recovered (7 days or less)	Percentage Recovered
1	96	82	85.4
2	83	69	83.1
3	91	73	80.2
4	47	35	74.5
5	60	42	70.0
6	27	22	81.5
7	69	52	75.4
8	72	57	79.2
9	89	76	85.4
10	46	37	80.4

a) $p = P(\text{Drug A} > \text{Drug B})$

$H_0: p = 0.5$

$H_a: p \neq 0.5$

Hospital	A	B	Signs
1	75.0	85.4	−
2	69.8	83.1	−
3	85.7	80.2	+
4	74.0	74.5	−
5	69.0	70.0	−
6	83.3	81.5	+
7	68.9	75.4	−
8	77.8	79.2	−
9	72.2	85.4	−
10	77.4	80.4	−

$n = 10$

$(-) = 8$

$(+) = 2$

→ $X \sim Bin(10, 0.5)$

$P(X \le 1) = 0.0107$

p-value $= 2(0.0107) = 0.0214$

RR: $x \le 1$ or $x \ge 9$

$x^* = 2 \not\le 1$, $2 \not\ge 9$

(Failed to reject H_0)

→ There is no sufficient evidence to conclude that one drug has higher recovery rate than other

b) Using Students t-test might be inappropriate because these datas are percentages/proportions & may not satisfy the normality assumption required for the t-test.

Also, sample size is small, $n = 10$

So, sign test is safer bcs it is nonparametric & do not require normality